This is a readme file generated by jp2020213248 for EBU6230.

It is about how to run the software and use the codes.

1. Installation:

download Matlab2021b;

unzip the jp2020213248 folder and load into Matlab2021b;

attention: when running the code, Matlab requires you to choose the current folder into "code" folder, where all codes are located.

2. User guidance:

attention: All file should be load via like:

path1 = '..\dataset\Lena512C_ASCII2014.ppm'.

Exercise 1(a)

[I,w,h,level] = BUPT_ppmRead(path)

path: path of ppm file

action example:

path = '..\dataset\Lena512C_ASCII2014.ppm';

BUPT_ppmRead(path);

[I,w,h,level] = BUPT_pgmRead(path)

path: path of pgm file

action example:

path = '..\dataset\Lena512_ASCII2014.pgm';

BUPT_pgmRead(path);

BUPT_ppmWrite(I,type,name)

I: the input matrix of ppm file

type: P3 or P6

name: file name

action example:

path = '..\dataset\Lena512C_ASCII2014.ppm';

[I,w,h,level] = BUPT_ppmRead(path);

BUPT_ppmWrite(I,'P3','write_ppm_result');

BUPT_pgmWrite(I,type,name)

I: the input matrix of pgm file

type: P2 or P5

name: file name

action example:

path = '..\dataset\Lena512_ASCII2014.pgm';

[I,w,h,level] = BUPT_pgmRead(path);

BUPT_ppmWrite(I,'P2','write_pgm_result');

Exercise 1(b)

BUPT_format_converter(path,arg)

path: path of ppm file ready to be processed

arg: whether converting from RGB to YUV or YUV to RGB

action example:

1. just convert to grey-scale image:

path = '..\dataset\Lena512C_ASCII2014.ppm';

BUPT_format_converter(path,1);

2. convert YUV back to RGB format:

path = '..\dataset\convert_to_YUV.ppm';

BUPT_format_converter(path,'inverse');

Exercise 1(c)

BUPT_sub_sample(path, vertial_rate, horizontal_rate)

path: the path of the pgm file

vertial_rate: rate for sub-sampling in vertial direction

horizontal_rate: rate for sub-sampling in horizontal direction

action example:

```
path = '..\dataset\Lena512_ASCII2014.pgm';
BUPT_sub_sample( path, 1, 2);
```

Exercise 2(a)

BUPT_quantization(path,factor)

path: the path of the pgm file

factor: quantisation factor

action example:

```
path = '..\dataset\Lena512_ASCII2014.pgm';
BUPT_quantization(path,2);
```

Exercise 2(b)

BUPT_histogram(path)

path: path of pgm file ready to be processed

action example:

path = '..\dataset\Lena512_ASCII2014.pgm';

BUPT_histogram(path);

Exercise 2(c)

BUPT_histogram_equalization(path)

path: path of pgm file ready to be processed

action example:

path = '..\dataset\Lena512_ASCII2014.pgm';

BUPT_histogram_equalization(path);

path2 = '..\dataset\histogram_equalization_l.pgm';

BUPT_histogram(path2);

Exercise 2(d)

BUPT_histogram_modeling(path1, path2)

path1: path of pgm file being modelled

path2: path of pgm file ready to be processed

action example:

path1 = '..\dataset\Lena512_ASCII2014.pgm';

path2 = '..\dataset\Peppers512_ASCII2014.pgm';

BUPT_histogram_modeling(path1, path2);

Exercise 3(a)

BUPT_PPMnegative(p)

p: path of ppm file

action example:

p = '..\dataset\Lena512C_ASCII2014.ppm';

BUPT_PPMnegative(p);

BUPT_PGMnegative(p)

p: path of pgm file

action example:

p = '..\dataset\Peppers512_ASCII2014.pgm';

BUPT_PGMnegative(p);

Exercise 3(b)

[image] = BUPT_transform(I, th1, arg, procedure)

I: input image matrix (read from file path)

th1: the angle for rotation

arg: the angle for skewing

procedure: doing which (row/skew) first

action example:

```
path = '..\dataset\test.pgm';
[I,w,h,level] = BUPT_pgmRead(path);
BUPT_transform( I, 20, 50, 's' ); //skew first
```

Exercise 4(a)

BUPT_addSaltNoise(path)

path: the path of the pgm file

action example:

```
path = '..\dataset\Baboon512_binary2014.pgm';
BUPT_addSaltNoise(path);
```

[image] = BUPT_addGaussianNoise(path,m,var)

path: the path of the pgm file

m: mean of the Gaussian

var: varianze of the Gaussian

action example:

path = '..\dataset\Baboon512_binary2014.pgm';

BUPT_addGaussianNoise(path,0,0.01);

Exercise 4(b)

BUPT_up(path, category, factor)

path: the path of the pgm file

category: 'n' for nearest neighbour, 'b' for bilinear, 'both' for showing both method of processing

factor: factor for up sampling

action example:

path = '..\dataset\Baboon512_binary2014.pgm';

BUPT_up(path, 'b', 3,6);

Exercise 5(a)

BUPT_lowpass(path, power, sigma)

path: path of pgm file

power: the power of gaussian noise

sigma: the standard variance of gaussian noise

action example:

path = '..\dataset\Lena512_binary2014.pgm';

BUPT_lowpass(path, 50, 0.5);

BUPT_rect(path,size)

path: the path of the pgm file

size: the size of filter

action example:

path = '..\dataset\Lena512_binary2014.pgm';

BUPT_rect(path,5);

Exercise 5(b)

BUPT_edge_ppm(path)

path: path of ppm file ready to be processed

action example:

path = '..\dataset\Lena512C_ASCII2014.ppm';

BUPT_edge_ppm(path);

BUPT_edge_pgm(path)

path: path of pgm file ready to be processed

action example:

path = '..\dataset\Lena512_binary2014.pgm';

BUPT_edge_pgm(path);

Exercise 6

BUPT_LoG_operater(path,sigma)

path: path of pgm file

sigma: the standard variance of gaussian noise

action example:

path = '..\dataset\Lena512_binary2014.pgm';

BUPT_LoG_operater(path,1);

BUPT_LoG_filter(path, sigma, size)

path: path of pgm file

sigma: the standard variance of gaussian noise

size: size of kernel

action example:

path = '..\dataset\Lena512_binary2014.pgm';

BUPT_LoG_filter(path, 3, 19);